

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. The exact finite object is the complete probability law on hard-sphere microstates, with a genuine polynomial algebra of factorial correlation coordinates. A normally summable full triangular BBGKY Duhamel corrector gives the Hamilton–Boltzmann logarithmic generator on bounded collision-increment cylinders. The kinetic state uses ordinary weak convergence with energy as a lower-semicontinuous containment function, so energy sublevels are compact without requiring second-moment convergence. Action compactness, bounded-core comparison, the B2 Laplace principle, and the perturbed-test theorem identify the unique nonlinear kinetic semigroup.

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1. ROUND-SIX POSITIVE CLOSURE: FULL-LAW MICROSCOPIC VALUES AND WEAK-ENERGY KINETIC SEMIGROUPS

The exact finite-volume object is the complete probability law on hard-sphere microstates. Correlation hierarchies are coordinates of that law, not random states and not a linear algebra. The microscopic logarithmic values are transported through the full law and converge, by the B2 joint LDP

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and an independent full-BBGKY perturbed-test construction, to the density–contact Lax–Oleinik semigroup. The kinetic state uses the ordinary weighted weak topology with energy as a lower-semicontinuous containment function; it does not require convergence of second moments.

1.1. Exact finite-volume law and observable algebra. Let Ω_ε be the disjoint union of all hard-core finite-particle phase spaces and let Φ_t^ε be the Liouville flow. The exact state space is

$$\mathfrak{P}_\varepsilon = \mathcal{P}(\Omega_\varepsilon), \quad \mathcal{U}_\varepsilon(t)\rho = (\Phi_t^\varepsilon)_\# \rho.$$

For a law ρ , denote its complete factorial correlation hierarchy by $G(\rho) = (g_k)_{k \geq 0}$. This is a one-to-one coordinate map on the exponential label class, but it is deterministic once ρ is fixed.

Let $\mathfrak{A}_{\alpha,\beta}$ be the completion of finite polynomials in factorial moment coordinates

$$\rho \longmapsto P(\langle F_1, g_{k_1}(\rho) \rangle, \dots, \langle F_m, g_{k_m}(\rho) \rangle)$$

for the norm obtained by summing absolute polynomial coefficients against the factorial hierarchy norm. This is a genuine commutative algebra because the product is the product of polynomials on law space; no false identity $g_{k+l} = g_k \otimes g_l$ is used.

Lemma 1.1 (Law-coordinate algebra). *The algebra $\mathfrak{A}_{\alpha,\beta}$ is closed under multiplication and under exponentiation of bounded density/contact cylinders. Composition with $\mathcal{U}_\varepsilon(t)$ is a strongly continuous algebra homomorphism from $\mathfrak{A}_{\alpha,\beta}$ to $\mathfrak{A}_{\alpha-Ct,\beta'}$ on the B2 short horizon.*

Proof. Polynomial multiplication is the Cauchy product of coefficient arrays, which is bounded by the defining absolute coefficient norm. A bounded cylinder has an absolutely convergent exponential power series in that Banach algebra. The full BBGKY semigroup transports every hierarchy coordinate of the exact law, with the factorial label-radius estimate proved in B2; substituting the transported coordinates into a polynomial preserves products. Density of finite polynomials and the hierarchy estimate extend the map and give strong continuity. $\square$

For a bounded microscopic terminal observable F_ε , define the exact logarithmic value at a law state

$$\mathcal{V}_{s,t}^\varepsilon(F_\varepsilon; \rho) = \frac{1}{\mu_\varepsilon} \log \int e^{\mu_\varepsilon F_\varepsilon(\Phi_{t-s}^\varepsilon Z)} \rho(dZ).$$

This is an exact log-Laplace functional on the complete law. It is not called a Markov semigroup on one-particle density. Conditional towers, when needed, are taken on the complete posterior law/history state.

1.2. The full triangular BBGKY corrector. Let a density/contact cylinder be

$$F(f, \Gamma) = \varphi(\langle f, \phi_1 \rangle, \dots, \langle f, \phi_m \rangle, \langle \Gamma, \psi_1 \rangle, \dots, \langle \Gamma, \psi_l \rangle),$$

where $\varphi \in C_b^3$, every ϕ_i is smooth with bounded collision increment, and every ψ_j is bounded continuous. Expand its microscopic exponential in connected factorial cumulants. Let $\mathcal{A}_\varepsilon^{\text{BBGKY}}$ denote the generator of the complete hierarchy, including all boundary maps $g_j \mapsto g_{j+1}$.

For each connected label number $j \geq 2$, let $D_j[F]$ be the uncanceled connected defect of the exponential generator after the lower levels have been corrected. Define the next corrector by the full upper-triangular Duhamel inverse

$$\mathbf{c}_{j+1}[F] = - \int_0^T \left(\mathcal{A}_\varepsilon^{\text{BBGKY}}(-s) D_j[F] \right)_{j+1} ds.$$

The subscript means the $(j+1)$ -label coordinate after the complete hierarchy, not an autonomous fixed-level propagator.

Lemma 1.2 (Normally summable connected correctors). *On the B2 short horizon there are $C_F < \infty$ and $0 < \rho < 1$ such that*

$$\|\mathbf{c}_j[F]\|_{\alpha, \beta}^{\text{conn}} \leq C_F \rho^{j-1}, \quad j \geq 2.$$

The corrected law functional

$$F_\varepsilon^{\text{corr}}(\rho) = F(g_1(\rho), \Gamma(\rho)) + \sum_{j \geq 2} \mu_\varepsilon^{1-j} \mathbf{c}_j[F](G(\rho))$$

converges absolutely in the full law-coordinate algebra and

$$\sup_{\rho \in \mathcal{K}} |F_\varepsilon^{\text{corr}}(\rho) - F(g_1(\rho), \Gamma(\rho))| \rightarrow 0$$

for every trace-hierarchy compact set $\mathcal{K}$.

Proof. The complete Duhamel hierarchy expands into time-ordered creation/contact trees. The B2 connected trace norm bounds the j -label coefficient by $j! C^j T^{j-1}$, while the factorial law-coordinate norm removes $j!$. Choose the common horizon so $CT < \rho < 1$. The recursive Duhamel definition cancels every connected defect at its first label number and transfers only the next triangular boundary term to the following level. Normal convergence permits reordering of the complete series. Since normalized correlations are bounded on $\mathcal{K}$, the factor μ_ε^{1-j} and the geometric label estimate make the corrector sum converge uniformly to zero. No finite hierarchy truncation is claimed to be monotone or to define a positive law. $\square$

Let $\mathcal{H}_\varepsilon$ be the logarithmic generator acting on the full law-coordinate exponential functional. The free and binary boundary terms at the first uncanceled level give

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \frac{1}{2} \int f f_* ((v - v_*) \cdot \omega)_+ (e^{\Delta p + \psi} - 1).$$

Theorem 1.3 (Full-BBGKY perturbed-test convergence). *For every cylinder in the bounded-increment core, the corrected tests of Lemma 1.2 belong to the graph domain of the complete BBGKY logarithmic generator and satisfy, locally uniformly on trace-hierarchy compact sets,*

$$\mathcal{H}_\varepsilon F_\varepsilon^{\text{corr}} \longrightarrow \mathbb{H} \left(f, \frac{\delta F}{\delta f}, \frac{\delta F}{\delta \Gamma} \right).$$

Both upper and lower perturbed-test inequalities hold for the same corrected sequence.

Proof. Apply the complete hierarchy generator term by term to the normally convergent corrector series. At each label number, the Duhamel equation cancels the full incoming defect, including the coupling to the next marginal. The only uncanceled first-order terms are free transport and the binary collision boundary. Exponentiation of the empirical cylinder changes the two-particle boundary value by $e^{\Delta p + \psi}$, which yields the displayed Hamiltonian. B2's cyclic sector contributes $O(\varepsilon^\alpha)$, and the geometric corrector tail contributes $O(\rho^K)$ before sending $K \rightarrow \infty$. The absolute graph norm permits the generator and limit to pass through the infinite series. The same exact corrected sequence yields both half-relaxed inequalities, so no order-preserving hierarchy truncation is used. $\square$

1.3. The compatible weak-energy state topology. Let $\mathcal{E}$ be the set of nonnegative densities with fixed mass and finite energy. Give it the ordinary weighted weak topology generated by bounded continuous tests and compactly supported smooth tests; energy is not required to converge. Put

$$\Upsilon(f) = \int (1 + |v|^2) f(dx, dv).$$

Then

$$\mathcal{E}_M = \{f : \Upsilon(f) \leq M\}$$

is weakly compact by tightness and lower semicontinuity. The referee's sequence with disappearing second moment converges in this topology and hence is not a counterexample to compactness.

Let $\mathcal{D}_b$ be the cylinder algebra generated by $\phi \in C_c^1$ and collision invariants, with the requirement $\|\Delta\phi\|_\infty < \infty$. This class is convergence determining for the weak topology. Every finite-coordinate doubling penalty built from it has bounded collision increment. Elasticity gives $\Delta|v|^2 = 0$, so the containment function is also compatible with the Hamiltonian.

Lemma 1.4 (Hamiltonian finiteness and continuity). *For $f, g \in \mathcal{E}_M$, bounded ψ , and $p, q \in \mathcal{D}_b$ in one bounded core set,*

$$|\mathbb{H}(f, p, \psi) - \mathbb{H}(g, q, \psi)| \leq \omega_M(d_w(f, g)) + C_M \|p - q\|_{C^1, \Delta},$$

where $\omega_M(r) \downarrow 0$ and $\|p\|_{C^1, \Delta}$ controls the transport derivative and bounded collision increment. Moreover

$$\mathbb{H}(f, D\Upsilon, 0) = 0$$

for the collision part and has the declared transport bound.

Proof. Bounded $\Delta p + \psi$ controls the exponential uniformly. The collision kernel grows only linearly in $|v| + |v_*|$. A uniform second-moment bound makes first moments uniformly integrable:

$$\int_{|v|>R} |v|f(dv) \leq M/R.$$

On a compact velocity ball, weak convergence makes $ff_*B(e^{\Delta p+\psi} - 1)$ continuous; the tail estimate gives the uniform modulus. Transport is handled identically. Energy is conserved at each elastic collision, giving the final assertion. $\square$

1.4. The additive action and kinetic semigroup. For balanced density–contact paths put

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \ell(d\Gamma/dA_f) dA_f,$$

with infinite value off the feasible set. Define

$$(S_{s,t}\Phi)(f_s) = \sup_{(f,\Gamma): f(s)=f_s} \{\Phi(f_t) - \mathcal{A}_{s,t}(f, \Gamma)\}.$$

Static preparation costs are not included in this transition action.

Lemma 1.5 (Compact action sublevels). *On an initial set with bounded mass and energy, every action sublevel is compact in the density weak-path/contact weak-measure topology. Concatenation preserves feasibility and adds the action.*

Proof. Energy is conserved by free transport and elastic contact balance, so all density marginals remain in one weakly compact $\mathcal{E}_M$. Entropy coercivity controls total contact mass and makes contact measures uniformly integrable relative to A_f . The balance equation bounds the time modulus of every test in a countable dense subset of $\mathcal{D}_b$. Prokhorov and Arzela–Ascoli/Jakubowski compactness give a convergent subsequence. B3’s closed balance theorem and lower semicontinuity of integral entropy preserve feasibility and the action. Splitting or joining a balanced path preserves all boundary traces and makes the entropy integral additive. $\square$

Theorem 1.6 (Kinetic Lax–Oleinik semigroup). *The operators $S_{s,t}$ are monotone, cash additive, preserve bounded uniformly continuous functions on energy sublevels, and satisfy*

$$S_{r,t} = S_{r,s}S_{s,t}.$$

For $\Phi \in \mathcal{D}_b$, the value is a viscosity solution of

$$\partial_s U + \mathbb{H}\left(f, \frac{\delta U}{\delta f}, 0\right) = 0, \quad U(T, f) = \Phi(f).$$

A bounded actual-contact reward enters as the ψ -argument.

Proof. Lemma 1.5 makes the variational supremum stable and attained after relaxation. Concatenation and splitting give dynamic programming. On a short interval, B3 Fenchel duality gives the viscosity upper inequality. The B2 normalized regular tilt supplies a recovery path for the lower inequality. Bounded contact work is already part of the same dual Hamiltonian. Stability under initial weak convergence follows from compact sublevels and regular recovery. $\square$

Theorem 1.7 (Comparison on the bounded-increment core). *Bounded upper-semicontinuous subsolutions and bounded lower-semicontinuous supersolutions on an energy sublevel, tested by $\mathcal{D}_b$ and the containment function Υ , satisfy comparison. Hence the value in Theorem 1.6 is the unique viscosity solution in this class.*

Proof. Choose a convergence-determining sequence $(\phi_j) \subset C_c^1$ with bounded collision increments. Double variables with

$$\frac{1}{2\eta} \sum_{j=1}^J 2^{-j} |\langle f - g, \phi_j \rangle|^2 + \delta(\Upsilon(f) + \Upsilon(g)) + \frac{|t - s|^2}{2\eta}.$$

Every derivative of the finite-coordinate penalty is a finite linear combination of the ϕ_j and stays in $\mathcal{D}_b$. The energy penalty has zero collision increment. Compactness of $\mathcal{E}_M$ gives maximizers. Apply the two viscosity inequalities and use Lemma 1.4; first let $\eta \downarrow 0$, then $J \rightarrow \infty$ and $\delta \downarrow 0$. A standard strict time perturbation contradicts a positive maximum. $\square$

1.5. Microscopic convergence. Let F_ε be a bounded density/contact cylinder and let the initial laws lie in one B2 exponentially compact trace-hierarchy class.

Theorem 1.8 (Microscopic log-Laplace convergence). *The exact finite-volume logarithmic values*

$$\frac{1}{\mu_\varepsilon} \log \mathbb{E}_{\rho_\varepsilon} \exp\{\mu_\varepsilon F_\varepsilon(\pi_t^\varepsilon, \Gamma_t^\varepsilon)\}$$

converge locally uniformly over regular initial densities to the variational kinetic value generated by $S_{s,t}$. The same limit is obtained by the full-BBGKY perturbed-test method of Theorem 1.3. Thus the density semigroup is a proved large-deviation limit, not an exact finite-density Markov closure.

Proof. The B2 good joint LDP and compact action sublevels give the Laplace principle, which is exactly the Lax–Oleinik variational value. Uniform exponential compactness makes the convergence local uniform on regular initial sets. Independently, Theorem 1.3, containment, and Theorem 1.7 identify the upper and lower half-relaxed limits with the same unique viscosity solution. Cylinder approximation extends the result to bounded uniformly continuous terminal values. $\square$

1.6. Static microcanonical preparation and Gaussian tangent. Non-conserved preparation moments constrain only the initial law. Actual mass, momentum, and energy invariants may additionally label dynamic components. Retain the B1 multiplier λ as a static coordinate of an augmented semigroup and optimize it once at the initial boundary; it is not reselected at every time block.

The B3 exactly centred Gaussian process, contracted by a finite collection of kinetic observables, has drift b and covariance A . Its entropic transform has generator

$$\mathfrak{G}^\theta F = b \cdot DF + \frac{1}{2} A : D^2 F + \frac{\theta}{2} DF^T A DF.$$

Theorem 1.9 (Round-six B4 closure). *The complete finite hard-sphere law has a well-typed observable algebra and full-BBGKY corrector calculus. Its logarithmic density/contact values converge to the unique weak-energy Lax–Oleinik semigroup on the bounded-collision-increment core. Initial preparation is performed once through the B1 static multiplier, and the risk-sensitive diffusion generator is the finite-dimensional contraction of the B3 exactly centred Gaussian process.*

Proof. The microscopic algebra and generator are Lemma 1.1 and Theorem 1.3. The kinetic semigroup and uniqueness are Theorems 1.6 and 1.7; microscopic convergence is Theorem 1.8. The B1 source-dependent conditioning formula and B3 Gaussian contraction give the final two assertions. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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